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Simulations of quantum neural networks

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Abstract

We explore by simulation ways in which an array of quantum dot molecules could serve as a quantum neural computer. First, we show that a single quantum dot molecule evolving in real time can act as a recurrent temporal quantum neural network. Inputs are prepared by fixing the initial states of a quantum dot molecule, and outputs determined by reading its value at a given time T later. The nodes of the network are the instantaneous states of the molecule at successive time slices. The nodes interact indirectly through their mutual interaction with local and phononic modes of the substrate. These modes can be preferentially excited optically, and, therefore, controlled externally. The number of excitations can thus be used as trainable “weight” parameters for a neural network. This network is shown to perform classical logic gates. By preparing the input state as a superposition state, multiple inputs can be encoded as a single initial state. Second, we simulate the possibility of a spatial, rather than temporal, design, as a Hopfield net. The network consists of a regular array of quantum dot molecules on a suitable substrate. The molecules interact indirectly as before, and, now, with each other directly through Coulombic interactions. Both of the quantum networks have none of the “wiring problems” of traditional neural nets: the necessary connections are supplied by the physical system itself. Computation is performed by the intrinsic physics of the physical system. The long range character of the phononic interactions takes the net beyond traditional local connectionist structures. The hypothesized increase in

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complexity and power, in going to the quantum regime, is demonstrated. We train the quantum Hopfield net using simultaneous recurrent backpropagation. © 2000 Elsevier Science Inc. All rights reserved.

1. Introduction

Artificial neural networks [1] can be viewed as parallel computational models made up of densely interconnected adaptive processing units. Due to their adaptive nature, they are extremely useful in applications in which algorithmic data are incomplete. Recently, there has been a resurgence of interest in quantum computers [2,3], because of their potential for being very much smaller and very much faster than classical computers, and because of their ability in principle to do heretofore impossible calculations. Clearly a neural computer which used the intrinsic power and parallelism of a quantum computer would be of great worth.

Quantum dot molecules [4,5] are nearby groups of islands of alien atoms deposited on a host substrate such as GaAs, see Fig. 1. Recent works [6,7] show that regularly distributed quantum dot arrays can be successfully fabricated. A group of four (or five) can be deposited in an arrangement like the dots on dice. If the individual dots are close enough to each other (on the order of 10–20 nm), then excess electrons deposited on the dots can tunnel between islands [8]. This leads, in the absence of environmental potentials, to a doubly degenerate ground state, corresponding to the electrons' being in opposite corners of the square. Alternatively, a group of two could be deposited, giving the more familiar dipole. These two states can be thought of as a qubit, the “up” and “down” states of a digital computer: a “two level system”, or TLS. Lent et al. [8–10] label these the “polarization” of the molecule, P ; we prefer S , the “spin” of the TLS, which we designate as ± 1 . The corresponding states in Dirac notation are $|+\rangle$ and $|-\rangle$.

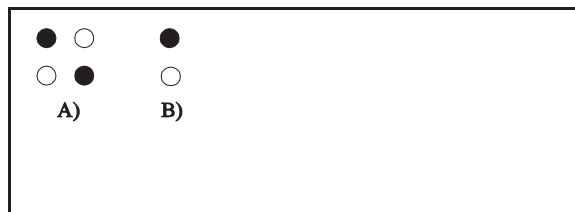


Fig. 1. Quantum dot molecules. In (A), two excess electrons preferentially occupy opposite corners in two equivalent ways; in (B), one excess electron can lie in either of the two equivalent dots. These states can be called $|+\rangle$ and $|-\rangle$ (a two state system, or TLS).

Lent et al. [8–10] have simulated arrays of quantum dot molecules as quantum cellular automata, in which the input molecule or molecules are constrained to be in one state or the other. The information propagates from one molecule in the array to the next through the Coulombic interactions of each pair, and the equilibrium (lowest energy state) of a molecule designated as the output is calculated. Suitably designed arrangements could in principle be constructed to perform basic logic gates. A different physical arrangement would have to be constructed for each desired gate.

2. Quantum temporal net

In previous works [11,12], we have shown by simulation that an array can instead act as a *neural* computer, with a single array performing *any* desired classical or quantum logic gate. Each dot molecule evolves in real time. The dot molecules at each time slice act as the “neurons” in this artificial neural network architecture. Again one or two molecules are designated as the inputs and their states fixed at a given time for any given computation, but now by manipulating the values of various environmental variables any desired logic gate output can be produced at the final or output time. In these earlier works, we fixed the states of two molecules at time zero (values for S) to be the input states, and read the value of one of them at a later time T to be the output. Thus to encode the logic gate OR, for example, we needed a set of four different inputs, $|+\rangle$ and $|-\rangle$ for each of two molecules, and trained the system to produce $|-\rangle$ for the input $|--\rangle$, and $|+\rangle$ for the other three possibilities. However, because this is a *quantum* system, the input states are not limited to being in the $|+\rangle$ and $|-\rangle$ states, but can be in a superposition, with any fraction of the probability amplitude in, say, the $|+\rangle$ state (the probability amplitude, up to a phase, of being in the $|-\rangle$ state is then fixed by normalization.) Thus even this simple system has in principle a continuum of input and output states (“quantum parallelism”). This is what we demonstrate in this paper, by using a single quantum dot molecule to compute logic gates, using superposition states as inputs.

To see how this system can be thought as a neural network, we note a mathematical isomorphism between the equation for the output of a neuron in a net and the expression for the time evolution of the quantum mechanical state of a system. This is because the state of the quantum system depends on its state at all previous times, just as the state of a neuron in a network depends on the states of all the neurons providing it input. For a traditional artificial neural network, the calculated activation of a φ_i of the i th neuron is performed on the signals $\{\varphi_j\}$ from the other neurons in the network, and is given by

$$\varphi_i = \sum_j w_{ij} f_j(\varphi_j), \quad (1)$$

where w_{ij} is the weight of the connection from the output of neuron j to neuron i , and f_j is a bounded differentiable real valued neuron activation function for neuron j . **Individual neurons are connected together into a network to process information from a set of input neurons to a set of output neurons.** The network is an operator $\mathcal{F}_W$ on the input vector $\boldsymbol{\varphi}_{\text{Input}}$. $\mathcal{F}_W$ depends on the neuron connectivity weight matrix W and propagates the information forward in “space” to calculate an output vector $\boldsymbol{\varphi}_{\text{output}}$

$$\boldsymbol{\varphi}_{\text{out}} = \mathcal{F}_W \boldsymbol{\varphi}_{\text{Input}}. \quad (2)$$

Similarly, we can write the expression for the time evolution of the quantum mechanical state of a system:

$$|\Psi(\varphi_f, T)\rangle = G_V(\varphi_f, T; \varphi_0, 0) |\Psi(\varphi_0, 0)\rangle, \quad (3)$$

where $|\Psi(\varphi_0, 0)\rangle$ is the input state, the initial state of the quantum system. $|\Psi(\varphi_f, T)\rangle$ is the output state, the state of the system at time $t = T$. G_V is the Green’s function, which depends on the potential created by the environment, V . G_V propagates the system forward in time, from initial state φ_0 at time $t = 0$ to final state φ_f at time $t = T$. This adjustable potential V contains the network weight parameters. This is equivalent to Eq. (4)

$$\begin{aligned} |\psi(\varphi_f, T)\rangle &= \int D\varphi(t) \exp \left\{ \frac{i}{\hbar} \int \left[\frac{m}{2} \dot{\varphi}^2(t) - V(\varphi, t) \right] dt \right\} |\psi(\varphi_0, 0)\rangle \\ &= \lim_{p \rightarrow \infty} \int d\varphi_1 \cdots d\varphi_p \exp \left\{ \frac{iT}{\hbar p} \sum_{j=0}^p \left[\frac{m}{2} \left(\frac{\varphi_{j+1} - \varphi_j}{T/p} \right)^2 \right. \right. \\ &\quad \left. \left. - V(\varphi_j) \right] \right\} |\psi(\varphi_0, 0)\rangle \end{aligned} \quad (4)$$

in which G_V is expressed as a Feynman path integral [13]. Qualitatively, the amplitude to arrive at state $|\Psi(\varphi_f, T)\rangle$ is equal to the sum over all possible paths $\varphi(t)$, of the amplitude of each of those paths. Each path is weighted by the complex exponential of the phase contributed by that path. This is given by the classical action for that path, m is the mass, $2\pi\hbar$ is Planck’s constant, and V is the potential energy. In the second line we express the path integral in its discretized form, with the number of discretization points, $p \rightarrow \infty$. If, instead, we take p to be finite, the p intermediate states can be considered to be the states of p quantum neurons, one at each time slice $j(T/p)$. (The weights have different values at different time slices; thus we can think of the timeline discretization points as virtual neurons, i.e., the neurons are distributed in time rather than in space.) The non-linearity necessary for neural computation is

inherent in the kinetic energy term, $(\varphi_{j+1} - \varphi_j)^2$, and in the exponential. Each of the p neurons' different possible states contributes to the final measured state. The *amount* it contributes can be adjusted by changing the potential energy, $V(\varphi)$. $V(\varphi)$ can thus be used as a trainable network weight value for the network training.

Specializing this to the TLS we consider here, the state of the system φ is the electronic state of the dot molecule, which we measure in the basis set $|\pm\rangle$, i.e., a TLS. The potential V is provided by the environment: the substrate phonons and an externally applied time varying external field. The phonons interact with the electronic state of the dot molecule. This phonon interaction can be represented by a set of Gaussians, i.e., the environment has a quadratic Hamiltonian, or, equivalently, a normal distribution. If the set is taken to be infinite, then any desired influence including dissipation can be produced [14]. Physically, the coupling would have to be weak enough to be represented accurately as linear. For example, GaAs substrate satisfies this, with a (unitless) electron–phonon coupling parameter of $0.08 \ll 1$ [15]. The phonon interaction provides additional trainable parameters for the network. To adjust the coupling in a physical implementation, we would optically excite more or fewer phonons at that particular frequency. Thus, the trainable weight parameters of the system are the numbers of phonons excited at a given frequency, the values of the external field at each time slice j , and the frequencies of the excitations. It should be noted that this system is not a feed-forward network: all virtual neurons are connected to all other virtual neurons, both forward and backward in time, by the effects of the environment. The path integral of Eq. (4) becomes

$$|\psi(T)\rangle = \lim_{p \rightarrow \infty} \int dx_1 \cdots dx_p \int d\sigma_{z1} \cdots d\sigma_{zp} \exp \left\{ \frac{iT}{\hbar p} \sum_{t=0}^p [K\sigma_x(t) + \Delta(t)\sigma_z(t) + \frac{m}{2}\dot{x}^2(t) + \frac{m\omega^2}{2}x^2(t) + \mu x(t)\sigma_z(t)] \right\} |\psi(0)\rangle, \quad (5)$$

where the two path integrations are over the states of the TLS and of the phonon, x (we include one phonon frequency only for simplicity; the generalization to m phonon modes is obvious). We use the Pauli matrix σ_z to represent the instantaneous state of the dipole, $\Delta(t)$ is the external field, μ the coupling between the electronic state of the TLS and the phonon, and K is the tunneling parameter, which comes from the kinetic energy term in Eq. (4). Since we are uninterested in the detailed dynamics of the phonons, and in order to simplify the computation, we integrate out the $\{x_i\}$ degrees of freedom, which gives us (truncating, now, to finite p)

$$\begin{aligned}
\langle \pm | \psi(T/p) \rangle &= [\cos(KT/p) \sin(KT/p)]^{p/2} \sum_{S_1=\pm 1} \sum_{S_2=\pm 1} \cdots \\
&\times \sum_{S_{N-1}=\pm 1} \left\{ \tan^{-1/2 \sum_{j=0}^{p-1} S_j S_{j+1}} i^{-1/2 \sum_{j=0}^{p-1} |S_j - S_{j+1}|} \exp \left[-\frac{i}{\hbar} \sum_{j=0}^p \Delta_j S_j \right] \right. \\
&\times \left. \exp \left[\sum_{j \neq j'=0}^p S_j E_{jj'} S_{j'} \right] \right\}, \quad (6)
\end{aligned}$$

where $\{S_j\}$ are the instantaneous values ± 1 assumed by the TLS at time slices $\{j\}$, and the effects of the substrate phonons are contained in the influence phase E , proportional to the response function and to the square of the coupling parameter μ . Multiplying on the left of Eq. (6) by $\langle \pm |$ (taking the projection) corresponds to performing the measurement of the state of the system at time T ; this is the output of the neural net. Fig. 2 shows a piece of the finite p system. The up and down arrows represent the state of the TLS at the time slices. Reality, of course, corresponds to infinite p ; thus, this neural network, in any physical implementation, has an infinite number of neurons, with an infinite connectivity. Finite number of time slices p can be thought of as corresponding to the time resolution of the changes in the external fields. Standard checks are of course performed to ensure that the discretization is fine enough that the calculation is correct.

We now set up a simulation of the quantum neural network to do a logic gate. We specify as inputs the initial ($t = 0$) state of a single quantum dot molecule: $|\psi(0)\rangle = \alpha|+\rangle + \alpha'|-\rangle$. α and α' can be any complex numbers, with the constraint that $|\alpha|^2 + |\alpha'|^2 = 1$ from normalization. Thus, we can in principle encode not just two but an infinite number of inputs! Here, however, we need only to be able to encode four inputs: those corresponding to (0,0), (0,1),

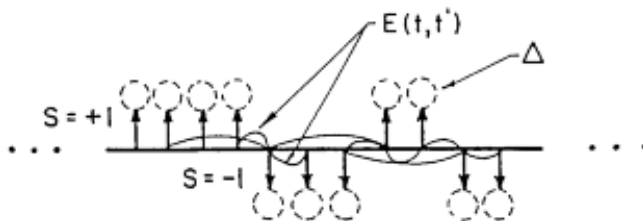


Fig. 2. Discretized representation of a single quantum dot molecule interacting with its environment. The instantaneous state of the system is represented by its spin state, $S = \pm 1$, and the integration over the environmental modes generates non-time-local interactions $E(t, t')$, some of which are shown here.

(1,0), and (1,1). We choose for these $\alpha = 0, 0.5, 0.5, 1$, respectively, with α' fixed by normalization. (We take the relative phase of the two components to be zero for simplicity.) For output we take the state of the molecule at the final time, which could be determined by measuring the external field of the quantum dot molecule, and threshold it at some value. We set the tunneling amplitude K to be the measured value [8] 0.03 meV (with internal molecular dot distance of 20 nm). We then train the net, using standard back-propagation in time [1] to minimize the error. By choosing appropriate values for the training rates η and the thresholding values, we have been able to train the net successfully to do classical logic gates. Figs. 3 and 4 show some representative results. For both logic gates, the temperature was 50 (meV)^{-1} and the total evolution time was

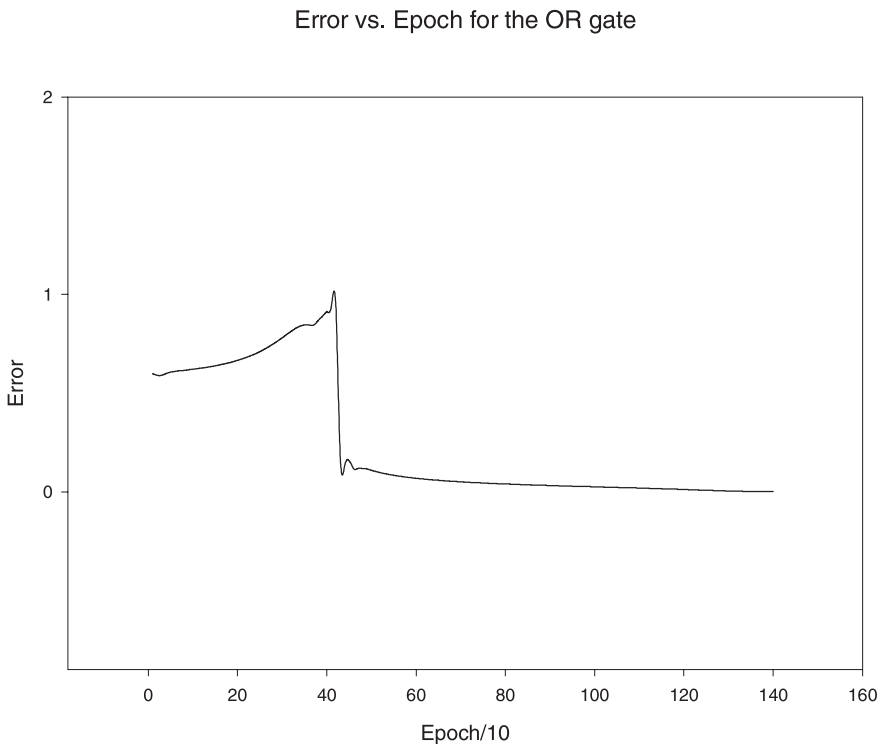


Fig. 3. Error as a function of epoch (number of passes through the training pairs) for the OR gate. The number of neurons was 4 and the number of phonon frequencies used was 1. The network trained the coupling constant (gross number of phonons excited times 0.08, the intrinsic coupling) from an initial value of 0.1 to a final value of 0.4336; the frequency was trained from 0.06 (initial guess at 2 K) to 0.01362. The time dependent field was trained from initial values of 1.0 at all time slices to final values of 0.983, 0.986, 0.989, 0.989. The final error was 0.01% after 1400 epochs.

Error vs. Epoch for the AND Gate

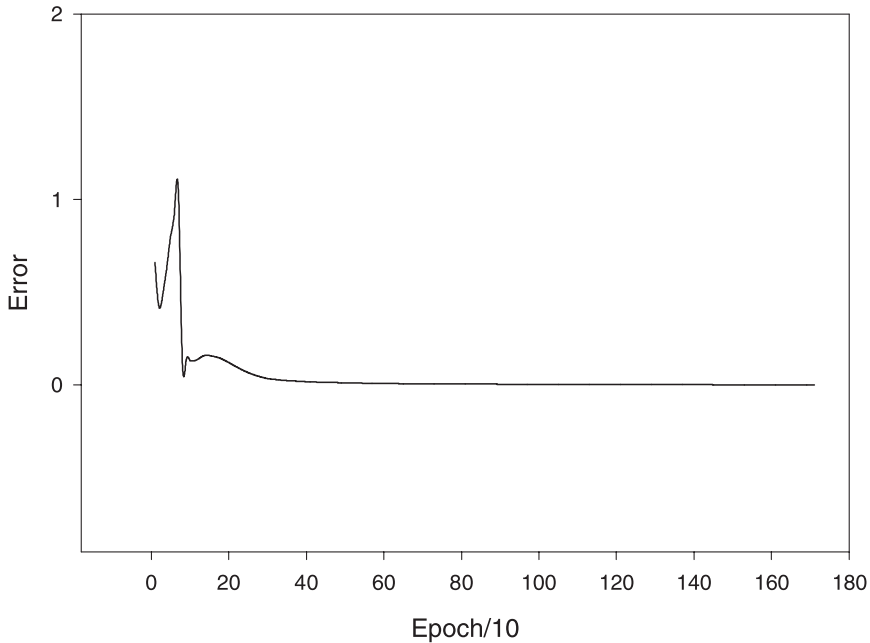


Fig. 4. Error as a function of epoch (number of passes through the training pairs) for the AND gate. The number of neurons was 5 and the number of phonon frequencies used was 2. The network trained the coupling constants (gross number of phonons excited times 0.08, the intrinsic coupling) from initial values of 0.1 and 0.4 to final values of 0.1845 and 0.8707; the frequencies were trained from 0.06 and 0.01 to 0.07263 and 0.09495. The time dependent field was trained from initial values of 1.0 at all time slices to final values of 0.65, 0.70, 0.69, 0.74, and 0.93. The final error was 0.1% after 1800 epochs.

100 ($\hbar/\text{meV}$). The net trained the number of phonons excited, the frequencies of those phonons, the external field at each time slice, and the temperature. Sensitivity to the temperature was, however, very small; very little change was observed in that variable. Thresholding was set to 0.5 (i.e., the output was considered a logical “1”, if the state were more likely than not to be in the $|+\rangle$ state, and a logical “0”, if the opposite were true). For these choices of initial values the OR gate was much more readily trained: the AND gate required an additional neuron and an additional phonon frequency. It should be noted that both gates can, of course, be trained from the *same* initial state with $n = 5$ and $m = 2$.

3. Quantum Hopfield net

For larger computations, pattern recognition and so on, it is clearly of interest to investigate the properties of a spatial array. Such an array can be thought of as a Hopfield net. In this formulation, each of the molecules corresponds to a node. We imagine preparing the initial states of each of the nodes. The network then evolves in time towards some final steady state. Again in the Feynman path integral formulation, the partition function is

$$\mathcal{Q} = \int \prod_{j=1}^N D\sigma_{zj}(t) \exp \left\{ - \int_0^\beta dt \sum_{j=1}^N K \sigma_{xj} + \int_0^\beta dt \sum_{j \neq j'=1}^N \gamma_{jj'} \sigma_{zj}(t) \sigma_{zj'}(t) \right. \\ \left. + \int_0^\beta dt \int_0^\beta dt' \sum_{j \neq j'=1}^N \sigma_{zj} E_{jj'}(t-t') \sigma_{zj'}(t') + \int_0^\beta dt \sum_{j=1}^N \Delta_j(t) \sigma_{zj}(t) \right\}, \quad (7)$$

where β is the inverse temperature in units of Boltzmann's constant, N is the number of quantum dot molecules, and all possible paths in imaginary time, $\sigma_z(t)$, are integrated over for each quantum dot molecule. The term containing σ_x represents the probability amplitude that the quantum dot molecule makes a transition from one state to the other (the “kinetic energy”). The second term is the Coulomb term; the third, the interactions with the phonons of the substrate, proportional to the auto-correlation function of the phonons; and the fourth, the external field term. The interactions that are non-local in time are generated by the interaction with the environment: physically, the dipole bumps its surroundings, which then as a result at a later (imaginary) time come back and bump the dipole. As before, we discretize the path integrals, though this time over the *temperature* loops (imaginary time) for each of the nodes, which gives us

$$\mathcal{Q} = \lim_{p \rightarrow \infty} \sum_{\{S_j(t) = \pm 1\}} \exp \left\{ - \frac{1}{2} \log(\tanh(\beta K/p)) \sum_{j=1}^N \sum_{t=0}^{p-1} S_j(t) S_j(t+1) \right. \\ \left. + \sum_{t=0}^{p-1} \sum_{j \neq j'=1}^N \gamma_{jj'} S_j(t) S_{j'}(t) + \sum_{t,t'=0}^{p-1} \sum_{j \neq j'=1}^N S_j(t) E_{jj'}(t-t') S_{j'}(t') \right. \\ \left. + \sum_{j=1}^N \sum_{t=0}^{p-1} \Delta_j(t) S_j(t) \right\}, \quad (8)$$

where the functions $\{S_j(t)\}$ are the instantaneous values assumed by each of the two-level systems j , at each imaginary time t ; i.e., the values of the operator σ_z in the up–down basis. This representation is exact in the limit that $p \rightarrow \infty$, and this corresponds to a very short imaginary time $\tau = -i\beta\hbar/p$. Classically, the

position or state of each TLS can be visualized as being represented by a point (either up or down, say); quantum mechanically, it can take on any admixture, and this is represented by the temperature loop (which in the classical limit shrinks to a single point), see Fig. 5.

We now draw the mathematical isomorphism. The contribution from a microstate to the partition function will be large if its associated energy is low. Thus, states that are more probable can be found by minimizing the negative of the argument of the exponential in Eq. (8). The argument can be rearranged into the form of the Hopfield energy or Lyapunov function [16,17]

$$L = -\frac{1}{2} \sum_{j \neq i} v_i T_{ij} v_j + \frac{1}{\lambda} \sum_i \frac{1}{R_i} \int_0^{v_i} g_i^{-1}(v) dv - \sum_i I_i v_i, \quad (9)$$

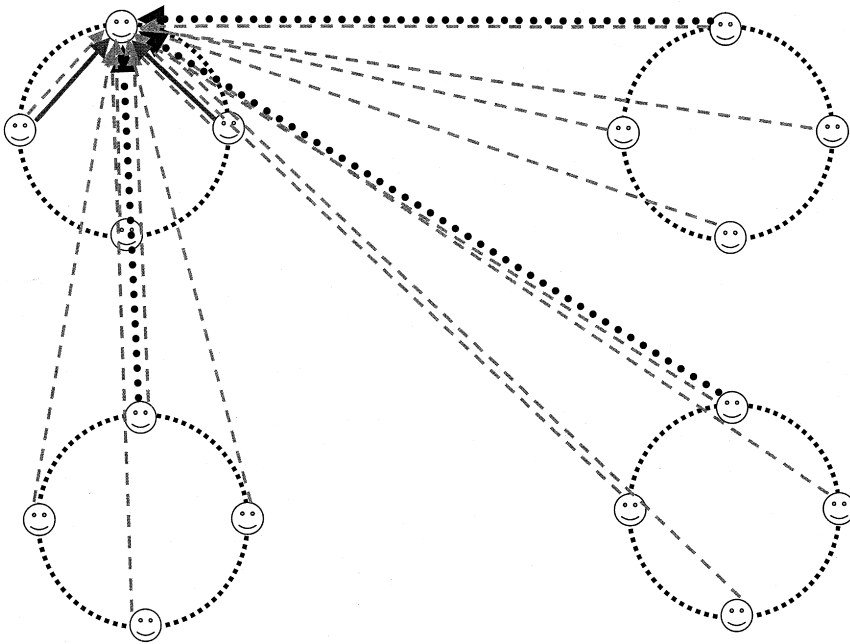


Fig. 5. Connectivity of the Hopfield net. Each closed loop represents a single quantum dot modedule (TLS). The smiling circles along each loop represent the state of the system at an instantaneous imaginary time τ , i.e., $S(\tau)$. The number of circles along each loop is $p = 4$; there are $j = 1, \dots, N = 4$ loops altogether. The more quantum mechanical the system, the floppier the loop. In the classical limit the loop shrinks to a single point. The Hamiltonian gives us interactions between each pair of circles: nearest neighbors are connected by the solid bonds (the “kinetic energy”); circles at equal imaginary times by dark dashed bonds (Coulomb); and each circle to each other circle by light dashed bonds, which come from having integrated out the phonons.

though what plays this role is not the energy of our system but its integral, or rather its discretized integral, over the temperature loop. That is, we make the correspondences

$$\begin{aligned} v_j &\leftrightarrow S_j(t), \\ T_{jj'} &\leftrightarrow \left\{ -\frac{1}{2} \log(\tanh(\beta K/p) \delta(t+1, t') \delta(j, j')) \right. \\ &\quad \left. - \frac{\beta}{p} \gamma_{jj'} \delta(t, t') + \left(\frac{\beta}{p} \right)^2 E_{jj'}(t - t') \right\}, \\ I_j &\leftrightarrow \Delta_j, \end{aligned} \quad (10)$$

where $v_i = g_i(\lambda u_i)$ is the sigmoid activation function of u , which we take to be $\tanh(\lambda u)$, and $\delta(k, k')$ is the Kronecker delta. We choose a sufficiently large gain parameter, λ , so that the second term becomes small. Each of the discretized states along each of the temperature loops for each of the molecules, corresponds to a node. In the continuum limit ($p \rightarrow \infty$) the number of nodes is infinite, and this is the quantum parallelism of the network. In the continuum limit the connectivity of the net is also infinite.

We now set up a simulation of the Hopfield quantum neural network. We choose the distance between neighboring molecules such that the effects of the tunneling amplitude, K , and of the Coulombic interactions, are approximately balanced. This is on the order of 50 nm if the internal distance (dot–dot distance) is taken to be 20 nm. We specify the initial states of each of an array of quantum dot molecules. We then solve the set of differential equations:

$$\frac{du_i}{dt} = \sum_j T_{ij} v_j - \frac{u_i}{R_i} + I_i, \quad (11)$$

whose steady-state solution minimizes L . The final converged state is taken as the output of the network.

The Hopfield net is a kind of auto-associative net. Given a set of physical parameters – in this case, the intra- and inter-molecular distances, the numbers and frequencies of phonons excited, the values for the external fields – the many-dimensional phase space has a set of local minima. If the net is initialized in a particular configuration, then it will equilibrate at one of these minima. Each local minimum is associated with a particular pattern, which can be said to be “stored” by the net’s physical parameters. This is one way of doing pattern recognition. Classically, the discrete net converges to the corners of an N -dimensional box, where N is the number of neurons. With the quantum Hopfield net the number of stable states is much larger, because of quantum superposition and because the connectivity is so much more complex. Another way of viewing this result is that the N -dimensionality, for a quantum Hopfield net, becomes p -fold larger, because each discretization dot along each

imaginary time loop can be thought of as a node; but this is somewhat misleading because the states of the discretization dots are not experimentally accessible.

To train the Hopfield net, we have developed a version of Werbos's [18] simultaneous recurrent back-propagation algorithm. The error at the k th dot molecule is defined to be

$$Error_k = \frac{1}{2} [d_k - S_k^{\text{avg}}]^2, \quad (12)$$

where S_k^{avg} is the output is taken as the average over the temperature loop for the k th dot molecule and d_k is the target desired value. The gradient of $Error_k$ with respect to an arbitrary weight parameter w for gradient descent training is

$$\frac{\partial Error_k}{\partial w} = [d_k - S_k^{\text{avg}}] \sum_t f'(u_k(t)) \frac{\partial u_k(t)}{\partial w}. \quad (13)$$

Initial errors $\tilde{\delta}_k(t)$ are assigned as initial states of each temperature point at the k th dot molecule as

$$\tilde{\delta}_k(t) = [d_k - S_k^{\text{avg}}] f'(u_k(t)) \quad (14)$$

and the system is allowed to converge again to an equilibrium state where the output states of each dot molecule and its temperature points are the errors used to update the weights according to the equation

$$w^{\text{new}} = w^{\text{old}} + \eta \tilde{\delta}_k(t) \left[\frac{\partial u_k(t)}{\partial w} \right]. \quad (15)$$

We compute the partial derivatives of each u needed above with respect to the adjustable parameters $\{\mu\}$, $\{\omega\}$, $\{I\}$, and the inverse temperature β , by differentiating Eq. (11). At equilibrium, the left hand side of Eq. (11) is zero, and the partial derivatives of each u can be calculated from

$$\frac{u_i}{R_i} = \sum_j T_{ij} v_j + I_i, \quad (16)$$

where the $\{T_{ij}\}$ contain the weight parameters according to the correspondence given in Eq. (10). Preliminary results show that we can in fact train logic gates using this algorithm.

Acknowledgements

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References

- [1] P.D. Wasserman, *Neural Computing Theory and Practice*, Van Nostrand Reinhold, New York, 1989.
- [2] A. Barenco, D. Deutsch, A. Ekert, R. Jozsa, *Phys. Rev. Lett.* 74 (1995) 4083.
- [3] J.J. Cirac, P. Zoller, *Phys. Rev. Lett.* 74 (1995) 4091.
- [4] M. Kemerink, L.W. Molencamp, *Appl. Phys. Lett.* 65 (1994) 1012.
- [5] R. Ugajin, *Phys. Rev. Lett.* 80 (1998) 572.
- [6] Y.H. Xie, S.B. Samavedam, M. Bulsara, T.A. Langdo, E.A. Fitzgerald, *Appl. Phys. Lett.* 71 (1998) 3567.
- [7] P. Petroff, *Bull. Am. Phys. Soc.* 45 (2000) 354.
- [8] G.L. Snider, A.O. Orlov, I. Amlani, X. Zuo, G.H. Bernstein, C.S. Lent, J.L. Merz, W. Porod, *J. Appl. Phys.* 85 (1999) 4283.
- [9] C.S. Lent, P.D. Tougaw, *Proc. IEEE* 85 (1997) 541.
- [10] C.S. Lent, P.D. Tougaw, W. Porod, in: *Proceedings of the Third Workshop on Physics and Computation (PhysComp 94)*, IEEE Computer Society Press, Silver Spring, 1994, p. 5.
- [11] E.C. Behrman, J. Niemel, J.E. Steck, S.R. Skinner, in: *Proceedings of the Fourth Workshop on Physics and Computation (PhysComp 96)*, New England Complex Systems Institute, 1996, p. 22.
- [12] E.C. Behrman, J.E. Steck, S.R. Skinner, in: *Proceedings of the Joint Conference on Information Science, Association for Intelligent Machinery*, 1998, p. 205.
- [13] R.P. Feynman, A.R. Hibbs, *Quantum Mechanics and Path Integrals*, McGraw-Hill, New York, 1965.
- [14] A.O. Caldeira, A.J. Leggett, *Ann. Phys.* 149 (1983) 374.
- [15] Y. Wan, G. Ortiz, P. Phillips, *Phys. Rev. Lett.* 75 (1995) 2879.
- [16] J.J. Hopfield, *Proc. Nat. Acad. Sci. USA* 81 (1984) 3088.
- [17] J.J. Hopfield, D.W. Tank, *Biol. Cybernet.* 52 (1985) 141.
- [18] P.J. Werbos, *Neurocontrol and supervised learning: and overview and evaluation*, *Handbook of Intelligent Control, Neural, Fuzzy, and Adaptive Approaches*, Van Nostrand Reinhold, NY, 1992, pp. 73–74 (Chapter 3).